
Careers in AI & Full Stack Development

Education, working life and career opportunities.

AUDIENCE

Classes IX to XII

VENUE

KV RK Puram, Sector 2

DEPARTMENT

DEPFE, NCERT, New Delhi

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Full Stack Software Engineer

Software Developer at Orangewood Labs
Noida, India

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B.Tech in Computer Science Engineering
Noida Institute of Engineering and Technology, 2020 to 2024

RoboGPT

Web applications that make robotic arm programming easier for non-technical users.

Robot-control interfaces

Real-time 3D visualisation and connected systems using gRPC and WebSockets.

From software to deployment

Python, TypeScript, React, Next.js and cloud infrastructure for real client projects.

Full Stack Developer

Develops both the client and server sides of a web application, providing an end-to-end service.

Client side

Build user-facing websites and interfaces that people interact with.

Server side

Develop the application logic and services behind the interface.

Databases

Work on projects involving data storage and connect it to the application.

Work environment and personal traits

Places and hours

- Employment in **IT companies**
- Typically **5 to 6 days a week**
- **8 to 9 hours per day**
- Shift systems may apply

Personality traits

- Good with computers
- Enjoy analysing problems and situations
- Prefer clear instructions to follow
- Comfortable communicating with others

Career options

- Work as an employee
- Pursue freelancing
- Work for your own company
- The page notes opportunities for people with disabilities

Fresher Developer → Full Stack Developer → Senior Full Stack Developer

Education and course fees

Pathway shown by NCERT

- Complete **10+2 in Science: Physics, Chemistry and Mathematics**
- Bachelor's degree: **B.Tech / B.Sc / BCA / B.E.** in Computer Science / Information System
- Or a bachelor's followed by **M.Tech / M.Sc / MCA / M.E.** in the same field

Course fee

₹55,000 – ₹10,00,000

- Approximate course-fee range in the supplied page
- Fees vary by institution; this is not labelled as an annual fee
- Check course duration and current admission requirements before enrolling

This replaces the earlier summary-sheet pathway. Individual programmes may have different eligibility rules.

Expected income and funding

Approximate monthly income

₹24,000 – ₹1,67,000

- **Per month**, as quoted by the supplied NCERT page
- Underlying source cited by NCERT: **PayScale**
- Indicative and subject to change; not a guaranteed starting salary

Scholarships and loans

- Engineering-specific and institute merit scholarships
- **National Scholarship Portal**: central, UGC / AICTE and state schemes
- **Vidya Lakshmi** education-loan portal, as referenced in the page
- Bank education loans and some state student-credit schemes; check eligibility and terms

The source names schemes including IndusInd Foundation, Baba Gurbachan Singh and Samsung Star Scholar. Some entries are dated 2022; verify current availability.

Artificial Intelligence Engineer

Uses AI and machine learning to develop applications and systems that help organisations operate more efficiently.

Definition summarised from the supplied NCERT page, p. 615.

- 01** **Myths and Educational Pathway**
Misconceptions and NCERT study routes
- 02** **Nature of Work**
Day to day tasks and tools
- 03** **Working Conditions and Attributes**
Environment and skills that help
- 04** **Career Progression and Salary**
Growth from entry to leadership
- 05** **Opportunities and Colleges**
Where to work and where to study
- 06** **Courses and Future Outlook**
Building employability for the long run

Myths about AI

Myth: AI is always correct

AI can generate convincing but incorrect answers. Testing, reliable sources and human review still matter.

Myth: AI will replace every job

AI can automate tasks, but its impact varies by role. Human judgement, responsibility and domain knowledge remain important.

Myth: AI engineering is just prompting

Building dependable AI systems also requires programming, data handling, evaluation, deployment and monitoring.

Myth: Only an AI degree leads into AI

The NCERT page lists several routes, including Computer Science, Mathematics, Statistics, IT and AI / ML study.

Routes into **AI engineering**

The NCERT pathway begins with 10+2 in Science: Physics, Chemistry and Mathematics.

Degree or diploma

- Bachelor's in **AI, Mathematics, Statistics, Computer Science or Information Technology**
- Or a **diploma in AI and Machine Learning**
- Another route: a bachelor's in Maths / Statistics / CS / IT, followed by an **AI-related online certificate**

Postgraduate route

- A bachelor's followed by a **master's in a relevant field**
- Examples include CS & Engineering, AI, ML, Robotics, Data Science and Data Engineering
- The page also includes allied fields such as Electronics, Signal Processing and Computational Linguistics

The page references national, state and institute entrance exams, including JEE Main / Advanced and WBJEE. Requirements vary by course; verify current eligibility and duration.

Responsibilities and tools

Practical examples of AI engineering work; responsibilities vary by team and product.

Core Responsibilities

- Design, train and fine tune **machine learning and deep learning** models
- Collect, clean and prepare data, then engineer features
- Deploy models into products and **monitor them in production**
- Work on NLP, computer vision, recommendations and large language models
- Write reliable, production grade code and run experiments

Tools and Technologies

- **Languages.** Python, SQL, sometimes C++ or Java
- **Frameworks.** PyTorch, TensorFlow, scikit learn, Hugging Face
- **Cloud and DevOps.** AWS, GCP, Azure, Docker, Git
- **Data.** Pandas, NumPy, Spark, vector databases
- Collaborates with data scientists, product and software teams

Work environment **and** personal traits

Hours and setup

- Typically an **office setup**
- Usually **8 hours per day**
- Longer hours may be needed depending on workload and deadlines

Teams and access

- Work alongside **data scientists and IT experts**
- Collaborate to develop applications and systems
- The source notes opportunities for people with disabilities

Personality traits

- Enjoy science
- Be creative
- Like using computers
- Have an interest in studying human behaviour

Scholarships and education loans

Scholarships

- Engineering-specific and institute merit scholarships
- Examples in the page: IndusInd Foundation, Baba Gurbachan Singh, Samsung Star Scholar, NTPC and Foundation for Excellence
- **National Scholarship Portal** for central, UGC / AICTE and state schemes

Loan options

- **Vidya Lakshmi**: the portal referenced for viewing, applying for and tracking education loans
- Bank education loans, subject to the lender's eligibility and terms
- Some states offer student-credit schemes; the page mentions West Bengal, Odisha and Bihar

Several scholarship entries in the source are dated 2022. Availability, deadlines and portal arrangements can change; verify current details before applying.

From entry level to leadership

Junior Engineer

Develop and test systems with guidance.

Senior Engineer

Take greater ownership of technical delivery.

Principal Engineer / Engineering Manager

Progress through technical leadership or people management.

Vice President

Lead broader engineering functions and priorities.

Chief Technology Officer

Guide technology strategy across the organisation.

NCERT's indicative progression, not a fixed promotion timeline. Role descriptions are presenter explanations.

NCERT's indicative figures

Approximate course fee

₹43,400 – ₹10,00,000

- Course-fee estimate in the supplied page
- Varies from institute to institute
- The page does not specify an annual fee; check the full fee structure before enrolling

Approximate monthly income

₹42,000 – ₹1,33,333

- **Per month**, as stated in the supplied page
- Indicative and subject to change
- Not a guaranteed starting salary or a stage-by-stage pay scale

Figures reproduced from the supplied NCERT material, not a current salary survey. Confirm current fees and compare live role-specific salary information.

Employment opportunities

The NCERT page identifies the following places of work, among others.

Software and tech companies

Develop AI-enabled applications and systems.

Research facilities

Work in government or private research settings.

Engineering institutes

Contribute to technical projects and research environments.

Institutions **listed by NCERT**

Examples from the NCERT page

- **Indian Institute of Science**, Bengaluru, Karnataka
- **University College of Engineering, Osmania University**, Hyderabad, Telangana
- **Indian Institute of Technology Hyderabad**, Telangana

Before choosing a programme

- The source describes the list as **indicative only**, not a ranking
- Check the current department, course level and curriculum
- Verify admissions, entrance exams, fees and duration on the institution's website
- Compare undergraduate and postgraduate options separately

The supplied screenshot cuts off the remaining institution list; only fully visible entries are reproduced here.

Building employability for the long run

Courses and Certifications

- DeepLearning.AI specialisations by Andrew Ng
- Google, AWS and Azure ML certifications
- Stanford CS231n for vision, CS224n for NLP
- **Kaggle** competitions and Kaggle Learn

Internships and Portfolio

- Research internships at IITs and IISc
- Google Summer of Code and industry internships
- Open source and a **GitHub project portfolio**
- Build things and share them publicly

Future Outlook

- Develop skills in **generative AI and applied ML**
- Keep programming and data fundamentals strong
- Study **responsible and ethical AI**
- Continue learning as tools and roles evolve

THANK YOU

Questions & discussion

AI engineering and full-stack development: education, skills and professional life.

Tushar Gaurav · Full Stack Software Engineer